PULLBACK ALONG THE COMPARISON FROM THE PRO-ETALE TO THE ETALE SITE IS AN EQUIVALENCE ON LOCALLY CONSTANT SHEAVES OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the splitting of the base into disjoint open pieces on which the trivializing module is constant, the passage to the torsor of isomorphisms with a constant sheaf, and the conclusion from the agreement of torsors between the etale and pro-etale topologies.

Editorial note. The label lemma-compare-locally-constant names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-compare-locally-constant –start 4337 –end 4388 –local-id 3588 returned RESULT=OK (L3588 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Openness of the image of an fpqc covering. Descent, the lemma that the image of an fpqc covering morphism is open. Supplies the openness of the images of the pieces of a pro-etale covering, used to cut X into the disjoint open pieces U_M.
- The sheaf of isomorphisms with a constant sheaf is constant. Modules on Sites, the lemma identifying the sheaf of isomorphisms between two locally constant sheaves with a constant sheaf. Supplies the identification of the sheaf G of automorphisms of the constant sheaf M with the constant sheaf on the group of module automorphisms of M.
- Agreement of torsors for the etale and pro-etale topologies. Same chapter, the lemma comparing the first cohomology with nonabelian coefficients for the pro-etale and the etale topology. Supplies the fact that the torsor of isomorphisms between the constant sheaf and F has sections etale locally.

Source reference. Original lemma 03N0: sources/stacks-chow/tags.txt, source lines 4768–4768. Prerequisite pointer invoked at proof line(s) 4369; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 099X: sources/stacks-proetale/pinned-original-source.tex, source lines 4226–4226. Prerequisite pointer invoked at proof line(s) 4384; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 093T: sources/stacks-chow/tags.txt, source lines 11526–11526. Prerequisite pointer invoked at proof line(s) 4379; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. Let Λ be a ring. The functor ϵ^{-1} defines an equivalence of categories*

$$\left\{\begin{array}{c}\textit{locally constant sheaves}\\ \textit{of } \Lambda\textit{-modules on } X_{\textit{étale}}\\ \textit{of finite presentation}\end{array}\right\} \longleftrightarrow \left\{\begin{array}{c}\textit{locally constant sheaves}\\ \textit{of } \Lambda\textit{-modules on } X_{\textit{pro-étale}}\\ \textit{of finite presentation}\end{array}\right\}$$

Proof. Let $\mathcal{F}$ be a locally constant sheaf of Λ-modules on $X_{pro\text{-}étale}$ of finite presentation. Choose a pro-étale covering $\{U_i \to X\}$ such that $\mathcal{F}|_{U_i}$ is constant, say $\mathcal{F}|_{U_i} \cong \underline{M_i}_{U_i}$. Observe that $U_i \times_X U_j$ is empty if M_i is not isomorphic to M_j. For each Λ-module M let $I_M = \{i \in I \mid M_i \cong M\}$. As pro-étale coverings are fpqc coverings and by Descent, Lemma 03N0 we see that $U_M = \bigcup_{i \in I_M} \mathrm{Im}(U_i \to X)$ is an open subset of X. Then $X = \coprod U_M$ is a disjoint open covering of X. We may replace X by U_M for some M and assume that $M_i = M$ for all i.

Consider the sheaf $\mathcal{I} = \mathit{Isom}(\underline{M}, \mathcal{F})$. This sheaf is a torsor for $\mathcal{G} = \mathit{Isom}(\underline{M}, \underline{M})$. By Modules on Sites, Lemma 093T we have $\mathcal{G} = \underline{G}$ where $G = \mathit{Isom}_\Lambda(M, M)$. Since torsors for the étale topology and the pro-étale topology agree by Lemma 099X it follows that $\mathcal{I}$ has sections étale locally on X. Thus $\mathcal{F}$ is étale locally a constant sheaf which is what we had to show. □